Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/02/2008
PI Name:	Wichman, Indrek Sven Ph.D.		
Project Title:	Investigation of Diffusion Flame Tip Instability in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	wichman@egr.msu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	517-353-9180
Organization Name:	Michigan State University		
PI Address 1:	Department of Mechanical Engineering		
PI Address 2:	2555 Engineering Building		
PI Web Page:			
City:	East Lansing	**State:**	MI
Zip Code:	48824-1226	**Congressional District:**	8
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	04/01/2004	**End Date:**	05/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	4
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date corrected to 5/30/2008 (from 1/04/2008) per PI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Olson, Sandra (NASA Glenn Research Center)		
Grant/Contract No.:	NNC04AA50A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of ATHINA is to examine the transition of flame fronts to flamelet spread. This occurs in low gravity near-limit conditions. In addition, for Fire Safety reasons it is necessary to examine the reverse transition, from flamelet spread to active flame spread. This might occur when a fresh quantity of air enters the region, and poses a serious fire hazard aboard spacecraft. ATHINA thus has a basic science component and a definite Fire Safety component. The objective of ATHINA is to conduct basic science and applied engineering research on the flame-to-flamelet (and vice-versa) transition process. Experimental, theoretical, and numerical means are to be employed to that end. Fire Safety in space has become a larger concern as NAS has become more mission-oriented. The significance of ATHINA appears in both its basic science implications (the flamelet regime being a troublesome		

	intermediate between flaming and quenching in which the flame very capably "adapts" to maximize its survival in adverse conditions), and in its engineering implications (being directly concerned with fire and flame spread in near-limit microgravity environments). The importance to Fire Safety is the largest element of significance in this project.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The device developed for ATHINA (earth-based), alternatingly called a "hele-Shaw" facility and a "Turing" facility (thus presently unnamed) is useful for assessing the fire hazard in tight spaces, such as narrow ducts and walls, where fires may be initiated.
Task Progress:	Theoretical Progress: Theoretical work was carried out on two projects: (1) The “Fire Dynamics Simulator” (FDS) code of NIST, adapted to simulate flame to flamelet transitions on burning thin samples; (2) The 2-D physics based simulation of the ignition-to-flame-spread-transition problem, which was carried out in order to explore fundamental unsteady flame spread processes; This work was carried out by MSU graduate student Mr. Y. Long and PI I. S. WIchman. For (1) the NIST Fire-Dynamics Simulator (FDS) code was adapted to conduct a transient 3-D simulation of the transition from flame fronts to flamelet fronts. The goal was to simulate the experiments and to describe the flamelet transition processes, including flamelet formation, flamelet bifurcation and extinction. Certain features of the flame-to-flamelet transition, including the eventual bifurcations and extinctions that occur in real tests, were relatively successfully simulated in the numerical work. We were able in the simulation to examine the detailed flow and concentration and reactivity patterns near the flamelets and to draw important conclusions about the flamelet breakup or bifurcation. Certain parts of flamelets have locally higher reactivity and these increases can be exaggerated under some conditions. Under suitable conditions, bifurcation occurs. When conditions are not suitable, the flamelet extinguishes or continues to spread. For (2) we have continued to examine a 2-D ignition and transition to steady flame spread. Here we have conducted an analytical analysis and supplemented that with numerical simulations for comparison. Of interest are comparisons to the results predicted for certain “macro” parameters that dictate overall results, and additional comparisons to various “micro” parameters that govern the details of the flame structure as it spreads. The 2-D basic physics simulation has enabled us to compare our results to theoretical predictions. We have written a paper that is in its final editing stage prior to being submitted to a journal. The work of (1) was summarized in the PhD thesis of Mr. Y. Long, which was entitled “A Numerical Study of Flame Spread over Thin Cellulosic Fuels in Microgravity” and which was successfully defended in December, 2007. Experimental Progress: A new test facility was designed by M.S. student K. Adkins during this research period. This facility is almost constructed and requires only a few finishing steps. We plan to use this facility both to examine simulated low-gravity flame spread and to examine flame spread in narrow gaps and spaces. We have given the facility the new name “The Narrow Channel Apparatus” or NCA. . Experimental Analysis Progress: We have prepared a lengthy paper describing some of the results, which was accepted for publication subject to minor revisions in Combustion and Flame. The title of the paper is “Flame Spread Over Thin Fuels in Actual and Simulated Microgravity Conditions,” by S.L. Olson, F.J. Miller, S. Jahangirian, and I. S. Wichman. A second paper on this subject, focused more on flame spread testing and the details of the operation of the NCA facility and based almost entirely on the M.S. thesis of S. Tanaya is under preparation for a journal such as Fire and Materials. The tentative title of this paper is “On the Uses of the Narrow Channel Apparatus in Evaluations of Flame Spread and Material Flammability,” by I. S. Wichman, S. L Olson, F. J. Miller and S. A. Tanaya.
Bibliography Type:	Description: (Last Updated: 10/16/2017)
Articles in Peer-reviewed Journals	Wichman IS, Olson SL, Miller FJ, Tanaya S. "Experimental evaluation of flame and flamelet spread over cellulosic materials using the narrow channel apparatus." Fire and Materials. 2013 Nov;37(7):503-19. http://dx.doi.org/10.1002/fam.2143 [Note reported originally in Dec 2008 as "On the Uses of the Narrow Channel Apparatus in Evaluations of Flame Spread and Material Flammability. To be submitted, 2009."] , Nov-2013
Articles in Peer-reviewed Journals	Long Y, Wichman IS. "Theoretical and numerical analysis of a spreading opposed-flow diffusion flame." Proceedings of the Royal Society A. 2009 Oct;465(2110):3209-38. https://doi.org/10.1098/rspa.2009.0152 [Note reported originally in Dec 2008 as 'To be submitted to Proceedings of the Royal Society A, 2009.'] , Oct-2009
Articles in Peer-reviewed Journals	Wichman IS, Olson SL, Miller FJ, Haraiharan A. "Fire in microgravity. Cover article." American Scientist. 2016 Jan-Feb;104(1):44. https://doi.org/10.1511/2016.118.44 , Jan-2016
Articles in Peer-reviewed Journals	Olson SL, Miller FJ, Jahangirian S, Wichman IS. "Flame spread over thin fuels in actual and simulated microgravity conditions." Combustion and Flame. 2009 Jun;156(6):1214-26. https://doi.org/10.1016/j.combustflame.2009.01.015 , Jun-2009
Dissertations and Theses	Long Y. "A Numerical Study of Flame Spread Over Thin Cellulosic Fuels in Microgravity." Ph.D. thesis, Michigan State University, December 2007. , Dec-2007
Dissertations and Theses	Adkins KJ. "Design of a New Narrow Channel Apparatus that Simulates Low Gravity Conditions for Producing Near Limit Flames." M.S. thesis, Michigan State University, Summer 2007. , Aug-2007
Journal/Magazine covers	Wichman IS, Olson SL, Miller FJ, Haraiharan A. "Cover in American Scientist for article, 'Fire in Microgravity.' " American Scientist. 2016 Jan-Feb;104(1):44. https://doi.org/10.1511/2016.118.44 , Jan-2016 , Jan-2016